

Element

At. No

Group 17 elements
Electronegativity

F

9

[He] $2s^2 2p^5$

Cl

17

[Ne] $3s^2 3p^5$

Br

35

[Ar] $3d^{10} 4s^2 4p^5$

I

53

[Kr] $4d^{10} 5s^2 5p^5$

At

85

[Xe] $4f^{14} 5d^{10} 6s^2 6p^5$

Ts

117

[Rn] $7s^2 7p^5$

G.E.C - $ns^2 np^5$

First four elements - Halogen (sea salt producer)

At - Radioactive

Cl - NaCl

I - NaIO₃

F₂ - superhalogen

Br - MgBr₂

Abundance order

$F_2 > Cl_2 > Br_2 > I_2$

Fluorapatite - CaF_2

Fluorapatite - $3(a_3(PO_4)_2 \cdot CaF_2)$

Cryolite

Na_3AlF_6

Rock salt - NaCl

Sylvine - KCl

Homelite - AgCl

Caralite - $KCl \cdot MgCl_2 \cdot 6H_2O$

Atomicity is 2

F_2, Br_2, Cl_2, I_2

Physical state

F_2, Cl_2 - gases

Br_2 - liquid

I_2 - solid

Atomic Radius

F_2 to I_2 ↑

I.P value

- high I.P

I.P F_2 to I_2 ↓

I_2 easily formed due to low I.P

E. Negativity $F_2 > Cl_2 > Br_2 > I_2$

Electron affinity $Cl_2 > F_2 > Br_2 > I_2$

F - small size Cl_2 F_2 Br_2
 $-349 kJ$ $-333 kJ$ $-325 kJ/mole$

There will some repulsions

blw added e⁻ and electrons present in fluorine atom.

Oxidation state F Cl Br I

"F" always exhibit "-1" O. state

Cl, Br, I $-1, +1, +3, +5, +7$ O. state

$Cl - 3s^2 3p^5$ $-1 (e^-) +1$

1st E. state $- 3s^2 3p^4 3d^1$

$\boxed{1\uparrow} \boxed{1\uparrow 1\uparrow 1\uparrow} \boxed{1\uparrow} \boxed{} \boxed{} \boxed{} \boxed{} +3$

2nd E. state $- 3s^2 3p^3 3d^2$

$\boxed{1\uparrow} \boxed{1\uparrow 1\uparrow 1\uparrow} \boxed{1\uparrow 1\uparrow} \boxed{} \boxed{} \boxed{} \boxed{} +5$

3rd E. state $- 3s^1 3p^3 3d^3$

$\boxed{1\uparrow} \boxed{1\uparrow 1\uparrow 1\uparrow} \boxed{1\uparrow 1\uparrow 1\uparrow} \boxed{} \boxed{} \boxed{} +7$

25/4/23

Colour

Halogen exhibit colours

F_2 - yellow

Cl_2 - Greenish yellow

Br_2 - Reddish brown

I_2 - violet

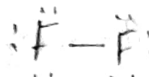
Halogens exhibit colours due to excitation and de excitation

Bond energy

Cl_2 - high B.E

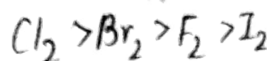
F_2 - has abnormally low B.E

F - small size

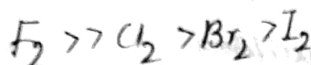


The repulsions b/w the lone pairs of F atoms weakens the bond b/w F & F.

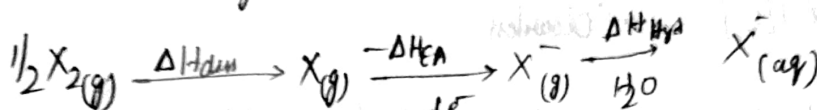
B-F order



Oxidising power



F_2 - strong oxidising agent
- High SRP value

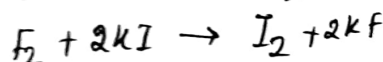
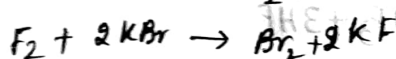
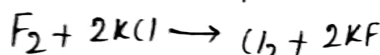


F_2 - low B.E, low E.A

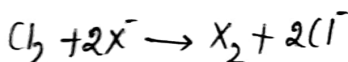
has high hydration enthalpy



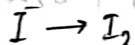
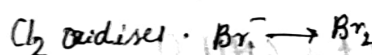
$X = Cl, Br, I$



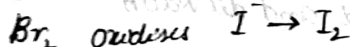
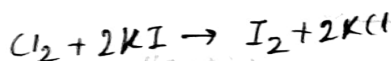
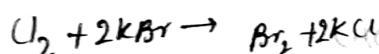
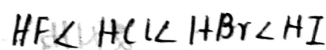
F_2 oxidises Cl^- to Cl_2 , Br^- to Br_2 , I^- to I_2



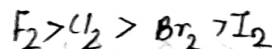
$X = Br, I$



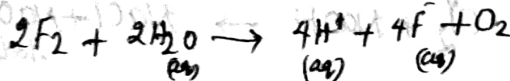
Reducing capacity of hydrides is

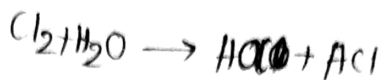
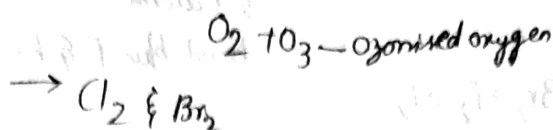
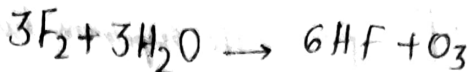


Reactivity:

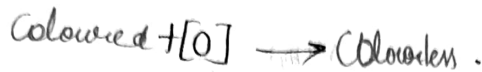
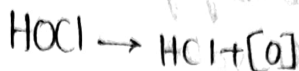


Solubility in water



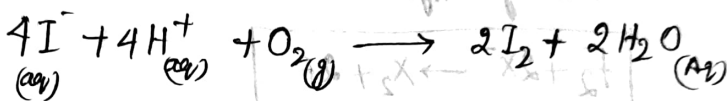


hypochlorous
acid



Cl_2 acts as bleaching agent by oxidation

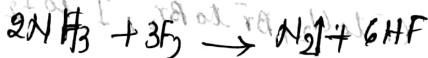
$\rightarrow$ Solubility of I_2 in water non spontaneous



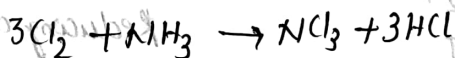
$\rightarrow$ with NH_3



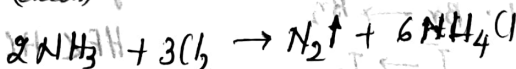
(excess)



(excess)



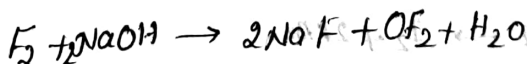
(excess)



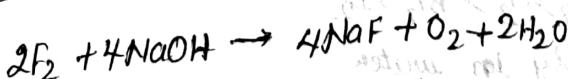
(excess)

$\rightarrow$ with alkalis ($NaOH$)

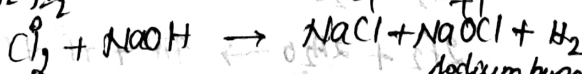
F_2 reacts with cold and dil $NaOH$



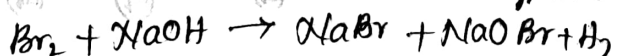
F_2 reacts with hot and con. $NaOH$



Cl_2, Br_2, I_2 reacts with cold and dil $NaOH$



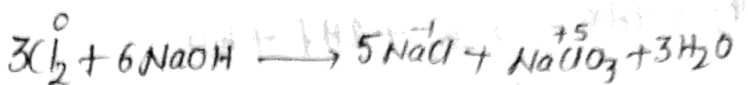
sodium hypochlorite





above are ^{sodium hypiodate} disproportionation rxns.

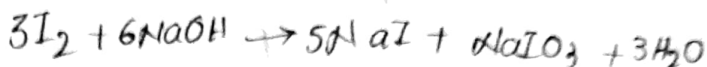
Cl_2, Br_2, I_2 reacts with hot and conc. NaOH



sodium chlorate



sodium bromate

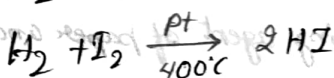
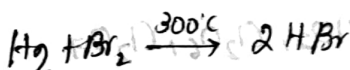
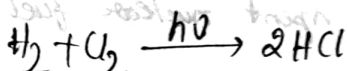
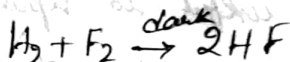


sodium iodate

with H_2

Hydrides

Reactivity $F_2 > Cl_2 > Br_2 > I_2$

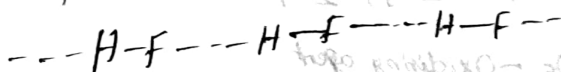


Thermal stability : $HF > HCl > HBr > HI$

B.P order : $HCl < HBr < HI < HF$

M.P order : $HI > HF > HBr > HCl$

HF has intermolecular H-bonds



Acidic nature

$HF < HCl < HBr < HI$

Conjugate base (or) Basic nature of halogen ions:

$F^- > Cl^- > Br^- > I^-$

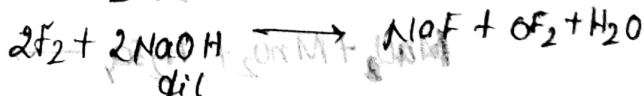
Oxides

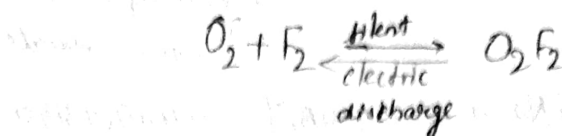
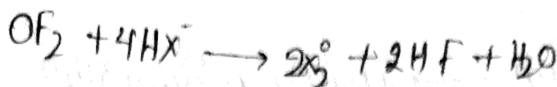
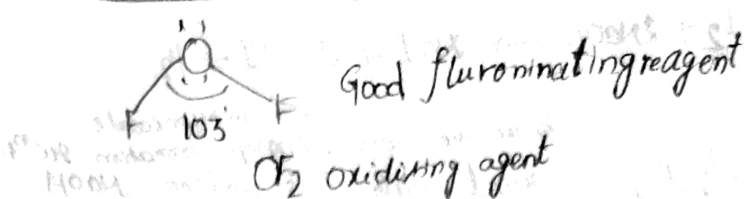
F_2 forms O_2F_2 and OF_2 (Oxygen fluorides)

OF_2 stable at $298K$

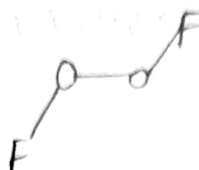
Room Temp.

O_2F_2 less stable





O_2F_2 has open book structure



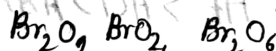
O_2F_2 oxidises "Pu" into PuF_6

Reaction is useful to separate "Pu" from "Pu ore" from spent nuclear fuel.

Cl_2 forms Cl_2O , ClO_2 , Cl_2O_6 , Cl_2O_7

Cl_2O — bleaching agent of paper and textile.

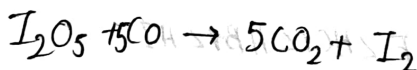
Br_2 forms unstable oxides



stable oxide is Br_2O_6

I_2 forms I_2O_4 , I_2O_5 , I_2O_7

I_2O_5 — oxidising agent



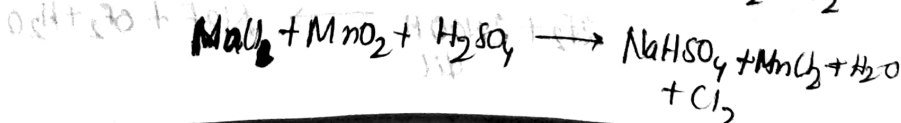
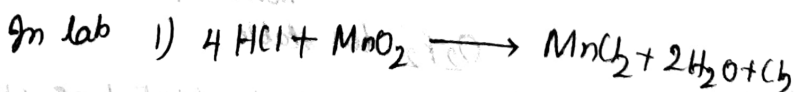
For estimation of "CO":

Stability order in $\text{I} > \text{Cl} > \text{Br}$
of oxides

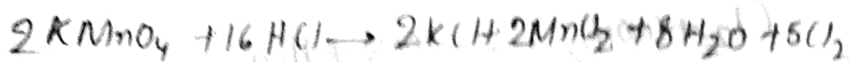
27/4/23

Chlorine (Cl_2)

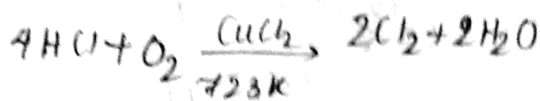
"Schale"



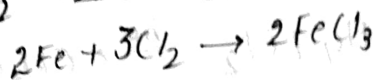
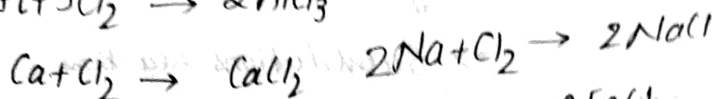
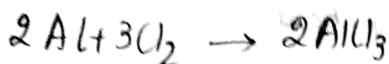
2) HCl is oxidised with KMnO_4



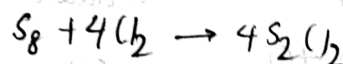
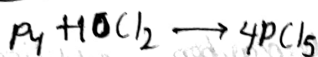
Deacon's process



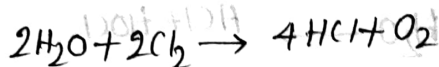
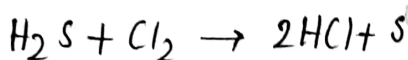
Cl_2 - greenish yellow gas, pungent smell, 2.5 times heavier than air
with metals



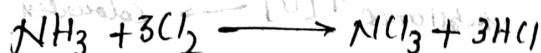
Non metals



Cl_2 has greater affinity towards hydrogen



→ with NH_3

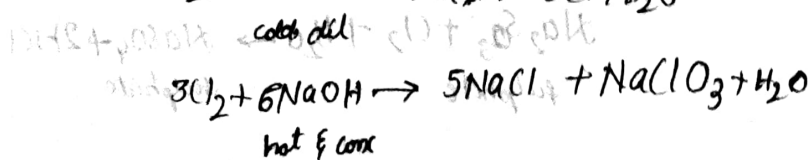
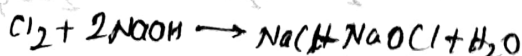


(Excess)

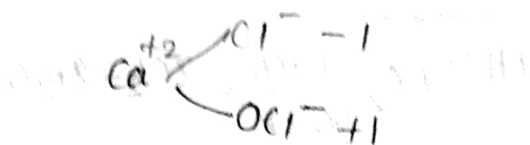
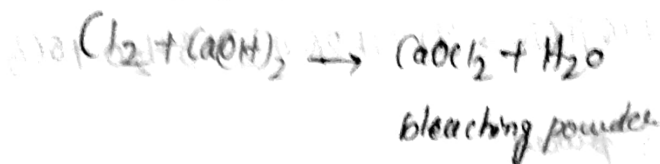


Excess

with NaOH or KOH



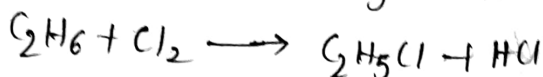
with $(\text{aOH})_2$



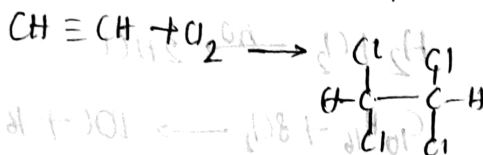
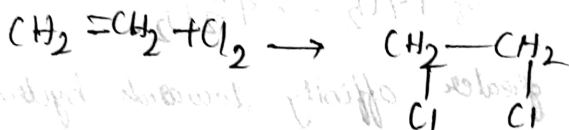
Composition of bleaching powder - $(\text{aOCl})_2 \cdot (\text{aH})_2 \cdot (\text{aOH})_2 \cdot 2\text{H}_2\text{O}$

Saturated hydrocarbons

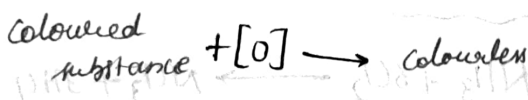
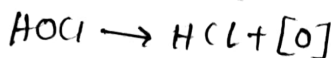
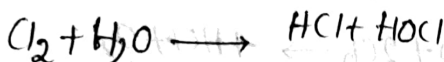
→ Substitutions reactions



Unsaturated hydrocarbon - addition reaction

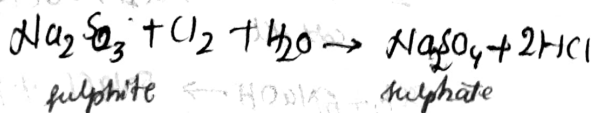
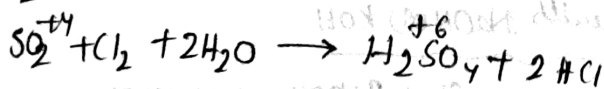
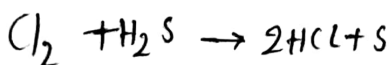


with H_2O



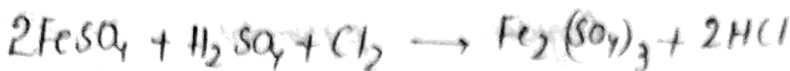
Cl_2 acts as bleaching agent by oxidation.

Cl_2 - oxidising agent

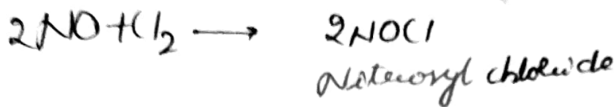
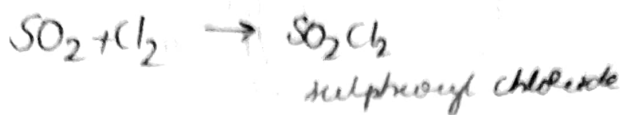




Sodium thiosulphate

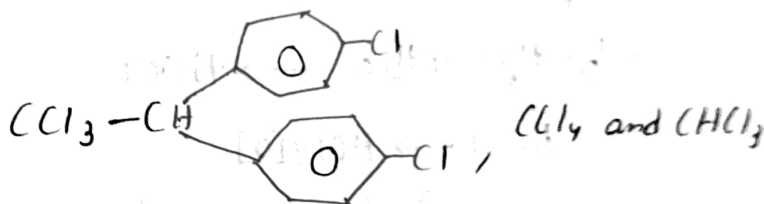


Addition reactions



Uses :

→ DDT



→ Bleaching agent of wood, paper, cotton and textile.

→ Extraction of Au & Pt.

→ Preparation of phosgene, tear gas, mustard gas.

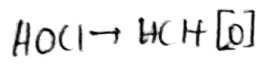
Oxoacids	F_2	Cl_2	Br_2	I_2
Hypohalous acid (I)	HO F hypofluorous acid	HO Cl hypochlorous acid	HO Br hypobromous acid	HO I hypoiodous acid
Halous acid (II)	—	HO ClO chlorous acid	—	—
Halic acid (III)	—	HO ClO_2 chloric acid	HO BrO_2 bromic acid	HO IO_2 iodic acid
Perhalic acid (VII)	—	HO ClO_4 perchloric acid	HO BrO_4 perbromic acid	HO IO_4 periodic acid
Acidic Nature		$HClO_3 > HBrO_3 > HIO_3$		
		$HClO_4 > HClO_3 > HClO_2 > HOCl$		

HOCl	HClO_2	HClO_3	HClO_4
ClO^-	ClO_2^-	ClO_3^-	ClO_4^-
hypochlorite ion	chlorite ion	chlorate ion	perchlorate

sp^3
linear

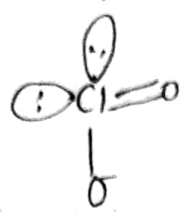


B. length $1.70 \text{ \AA}$



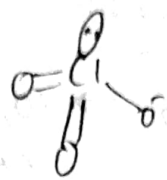
Good oxidizing agent

sp^3
Angular



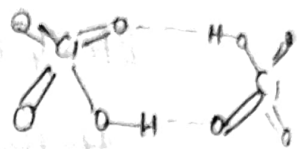
B. length $1.64 \text{ \AA}$

sp^3
pyramidal
1 LP



B. length $1.57 \text{ \AA}$

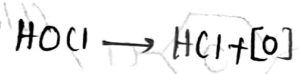
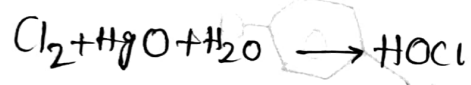
sp^3



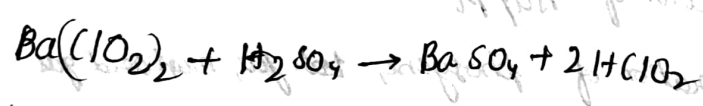
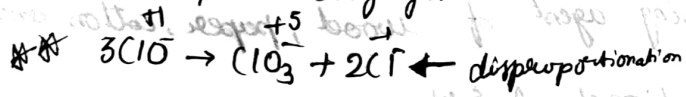
B. length $1.42 \text{ \AA}$

28/9/23

Preparation

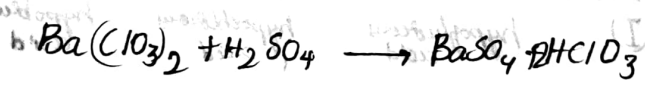
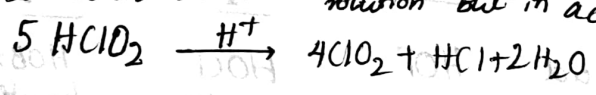


Good oxidizing agent

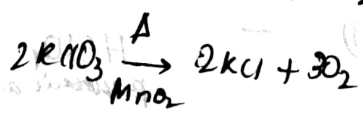
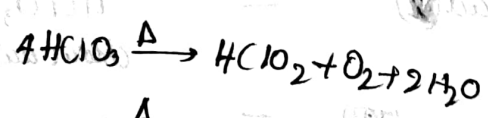


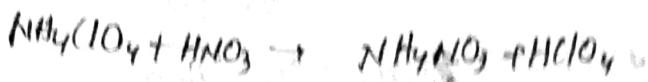
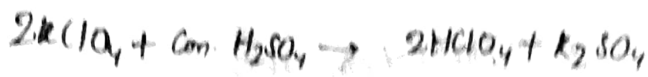
Barium chlorite

Stable in alkaline solution but in acidic soln

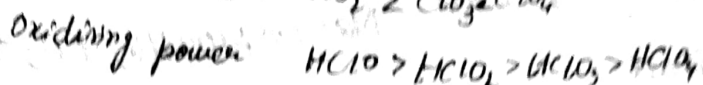
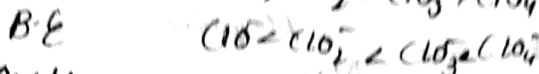
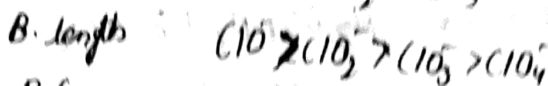
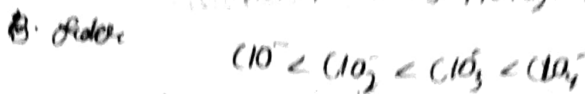
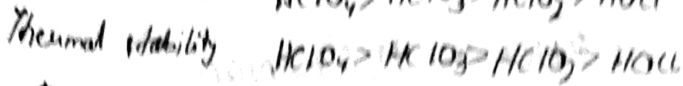
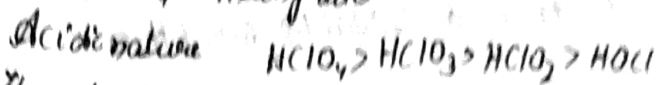


used in fireworks,
safety matches





$HClO_4$ - Strong acid



HCl Hydrochloric acid

"Glauber's"

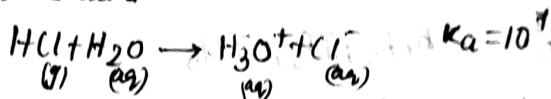


coloured gas

it can be derived by H_2SO_4

F.P - 159K B.P - 189K, colourless liquid

HCl - strong acid

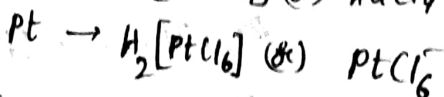
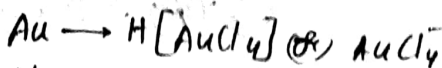
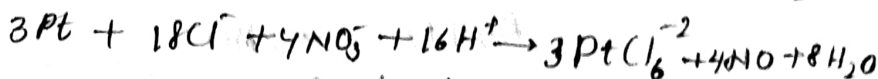
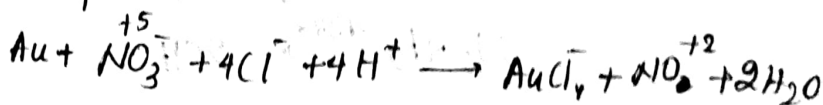


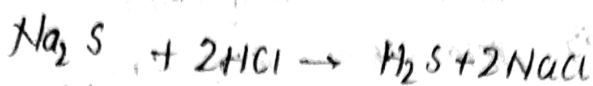
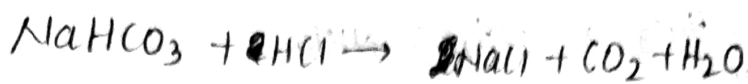
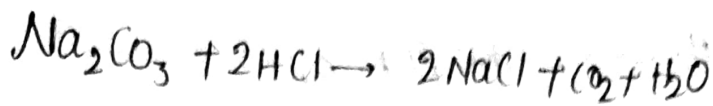
white fumes

Aqua regia

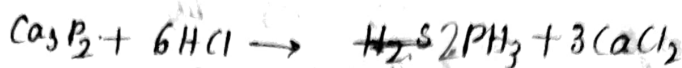
3 parts of HCl + 1 part of HNO_3

Au, Pt noble metals

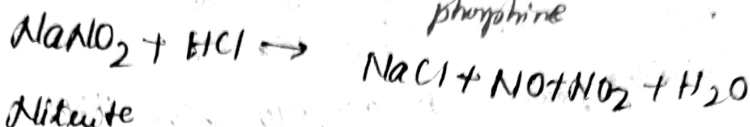




sulphide



phosphine

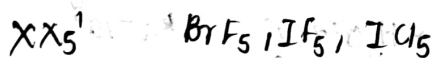
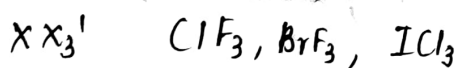
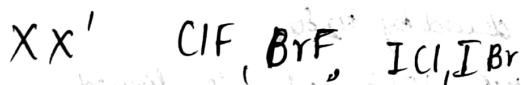


Nitrite

Uses:

1. Extraction of glue from bones.
2. HCl is used for the purification of bone black.
3. Preparation of Cl_2 .

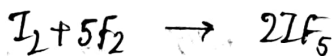
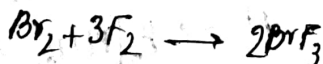
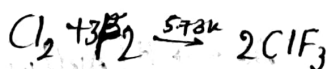
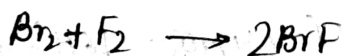
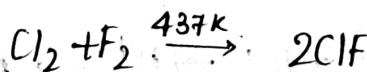
Interhalogen compounds

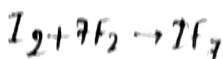
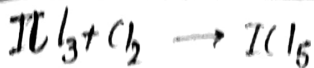


X - high electropositive halogen

X' - lower halogen

No. of atoms in interhalogen depends on Radius ratio





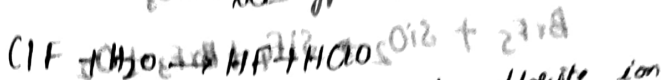
Covalent, diamagnetic, volatile liquids
(Br) solids except ClF - gas
M.P & B.P higher than halogen

Interhalogen compounds are more reactive than halogen (except F₂)

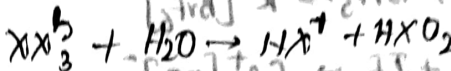
Hydrolysis



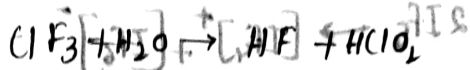
HOXO - hypohalite ion



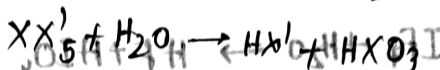
ClO⁻ - hypochlorite ion



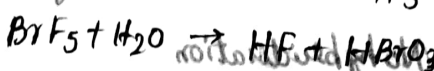
XO⁻ - halite ion



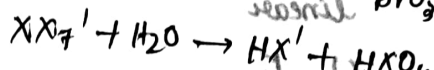
ClO₂⁻ - chlorite ion



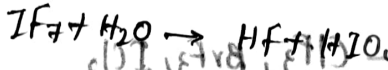
XO₃⁻ - halate ion



BrO₃⁻ - bromate ion

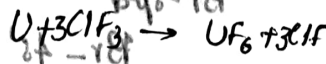


perhalate ion

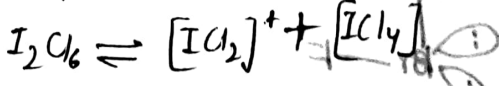
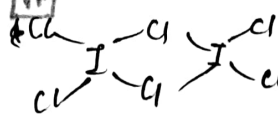


periodic acid

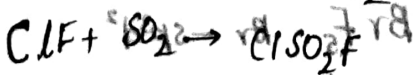
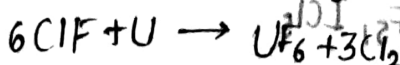
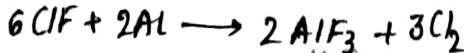
ClF₃ - good fluorinating agent



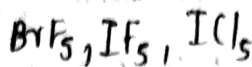
ICl₃ - exists as dimer I₂Cl₆



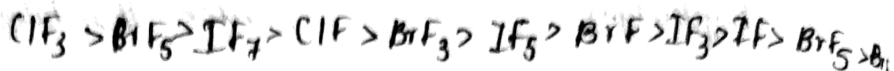
ClF - fluorinating agent



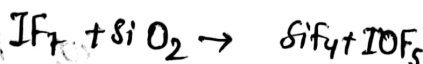
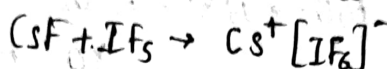
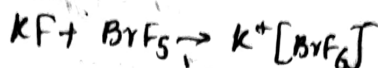
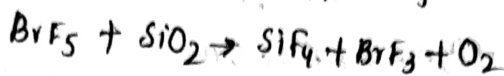
29/4/23



Reactivity order $\text{ClF}_n > \text{BrF}_n > \text{IF}_n$



more stable



XX' type - No hybridisation

Linear



XX'_3 type - $\text{ClF}_3, \text{BrF}_3, \text{ICl}_3$

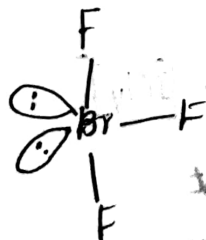
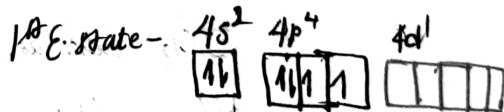


$\text{Br}-\text{sp}^3\text{d}$

$\text{Br}-4s^1 4p^5$

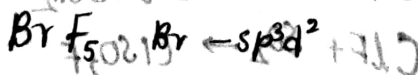
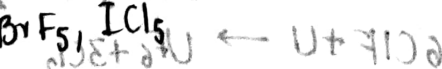
$\rightarrow 2 \text{ L.P}$

$\rightarrow \text{T-shape}$

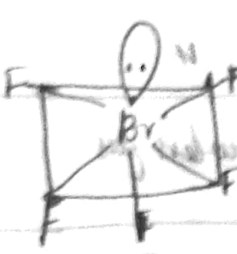


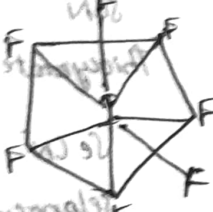
trigonal bipyramidal - 51)

XX'_5 - $\text{ClF}_5, \text{BrF}_5, \text{ICl}_5$



2nd E. state

$4s^2$ $4p^3$ $4d^2$
 $1 \cdot LP$ 111 11
 Square pyramidal

 sp^3d^2
 211
 211
 211
 111

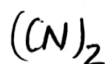
$XX'_7 - IF_7$
 sp^3d^3
 3^{rd} E. state
 No LP
 Pentagonal bipyramidal

 negative charge
 negative charge
 negative charge
 negative charge
 negative charge
 negative charge

Type	Formulae	Physical state and colour	Structure
XX'_1	ClF	colourless gas	Linear
	BrF	pale brown gas	"
	IF	detected spectroscopically gas	"
	$BrCl$	gas	"
	ICl	subby solid (2 form)	"
	IBr	black solid (0-1)	"
XX'_3	ClF_3	colourless gas	Bent-T shaped
	BrF_3	yellow green liquid	Bent-T shape
	IF_3	yellow powder	"
	ICl_3	orange solid	"

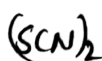
Type	Formula	Physical state and colour	Structure
AX_5	IF_5 BrF_5 ClF_5	colourless gas colourless liquid "	square pyramidal square pyramidal "
AX_7	IF_7	colourless gas	pentagonal bipyramidal

Pseudohalogens

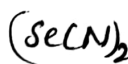
pseudohalogens



Cyanogen

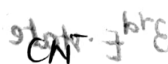


Thiocyanogen



Selenocyanogen

pseudohalide ions



Cyanate ion

labimaseffid larofathog

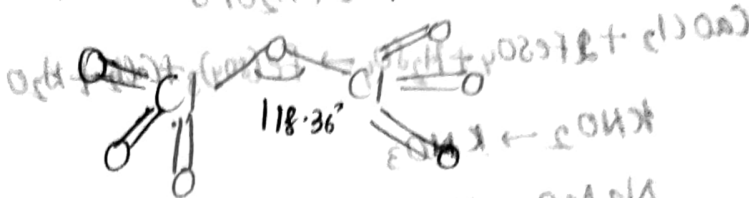
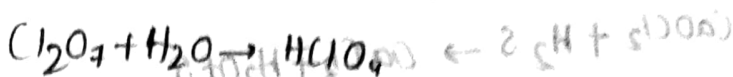


Thiocyanate ion

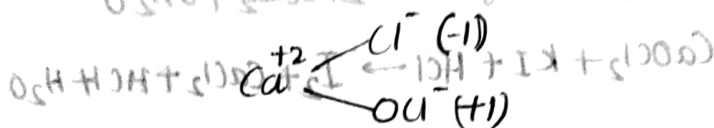


Selenocyanate ion

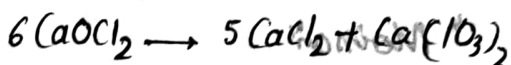
Structure	Formula	Type
Cl_2O	azide ion	XX'
ClO_2	ClO_3^-	XX_3
$2ClO_3^- \rightarrow Cl_2O_6$		



Bleaching powder - CaOCl_2



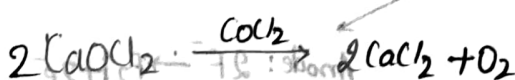
coloured amorphous powder



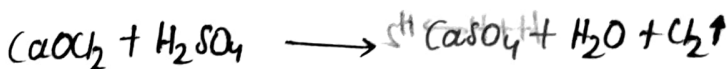
bottom: yellow solid calcium chlorate

In cold water - solution consists Ca^{+2} , Cl^- and ClO^- hypochlorite ion

In hot water - solution consists Ca^{+2} , Cl^- and ClO_3^-

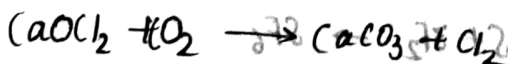


Insufficient amount



sufficient

available gas

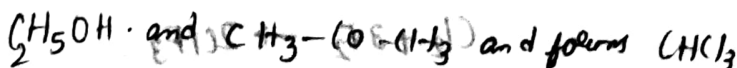


Good sample of bleaching powder

35-38% available chlorine

% of Cl_2 - 56%

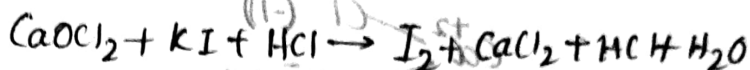
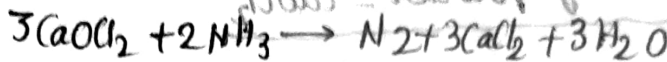
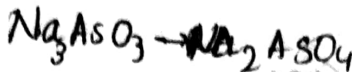
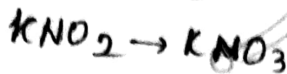
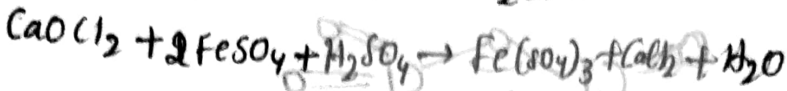
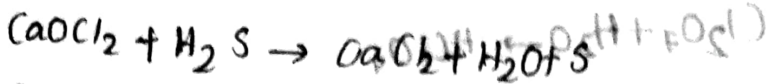
CaOCl_2 - bleaching agent



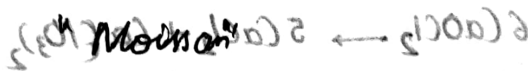
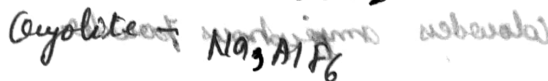
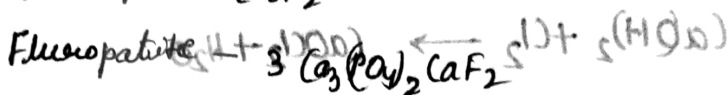
Chloroform

CaOCl_2 - Good oxidizing agent

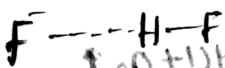
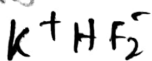
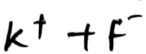
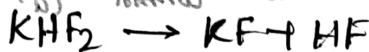
F_2



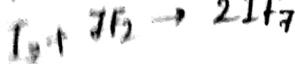
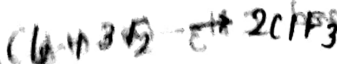
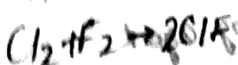
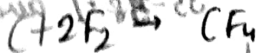
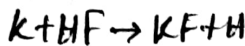
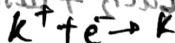
F_2 : fluorine - CaF_2

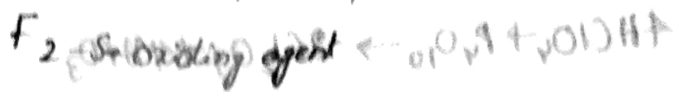
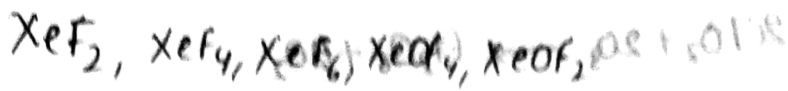


Whyte's method

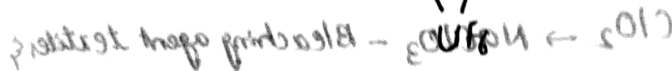
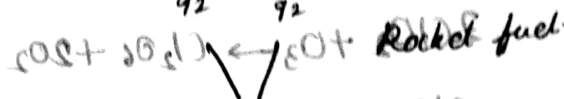
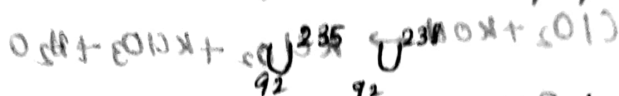
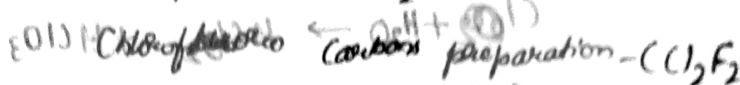
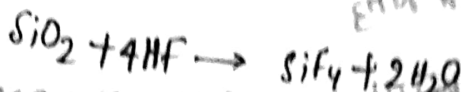


Cathode

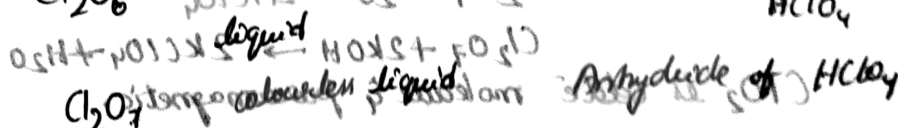
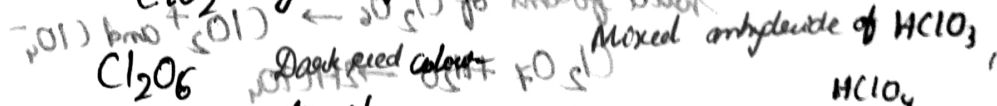
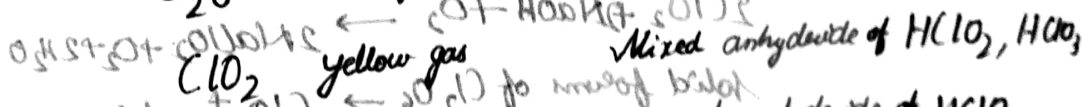
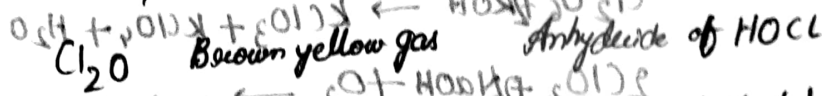
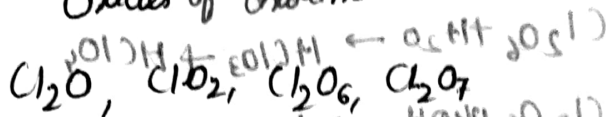




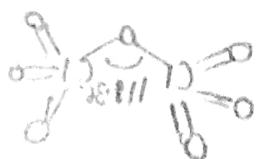
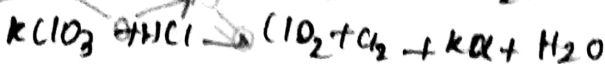
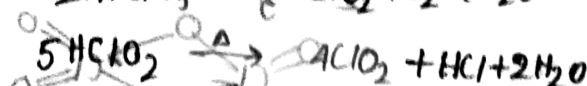
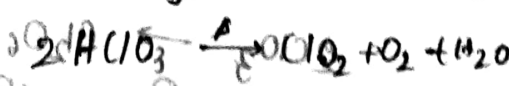
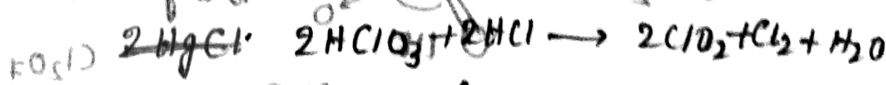
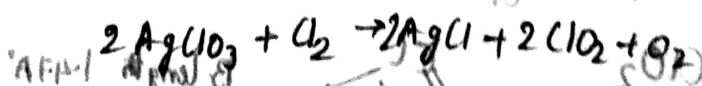
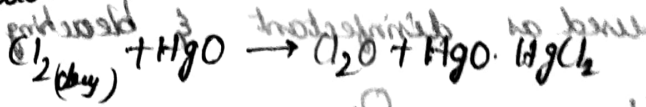
Silica in HF

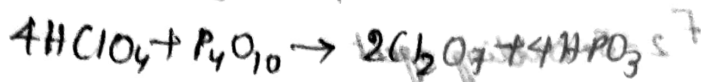
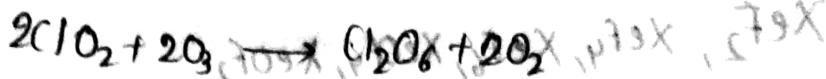


Oxides of Chlorine

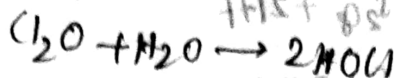


Preparation

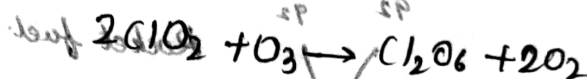
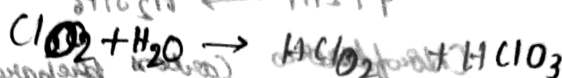
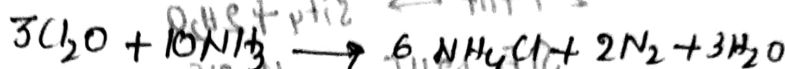




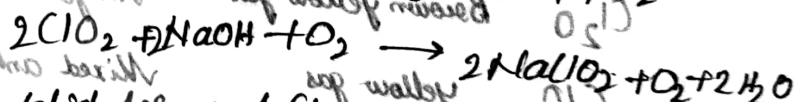
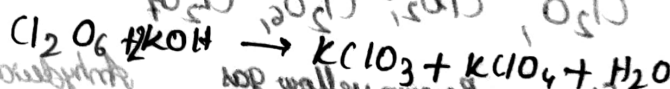
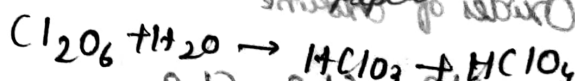
Properties



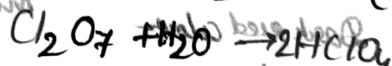
Cl_2O exploded in the presence of strong reducing agent NH_3



$\text{ClO}_2 \rightarrow \text{NaClO}_3$ - Bleaching agent textiles & paper

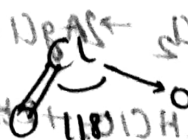
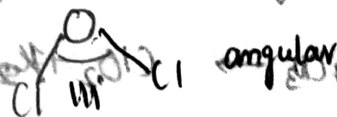
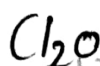


Solid forms of $\text{Cl}_2\text{O}_6 \rightarrow \text{ClO}_2^+$ and ClO_4^-



ClO_2 is odd e molecule & paramagnetic

Mixture ClO_2 and Cl_2 is euclidean mixture, used as disinfectant & bleaching agent.



B length 1.47 Å

